Chance Constrained Optimization with Kernel Density Estimation

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1 The Problem of Optimization Under Uncertainty

Every optimization model is a promise. It promises that if the decision it recommends is implemented, the constraints of the system will hold and the objective will be as good as the mathematics claims. That promise is honest only if the numbers inside the model are honest, and in almost every real deployment they are not. The demand a grid operator schedules against is a forecast, not a fact. The thrust a lander commands is corrupted by disturbances that sensors resolve only imperfectly. The returns a portfolio is built on are realizations of a process that is not available in closed form. This chapter establishes precisely what goes wrong when uncertainty is ignored, why the two classical remedies, ignoring it and armoring against the worst case, each have important limitations in the settings this thesis targets, and why the chance constraint, a constraint required to hold with a prescribed probability, offers a principled middle ground between them. It then exposes the three structural reasons chance constraints are computationally hard, builds the case for estimating the relevant probability distributions from data rather than assuming them, and introduces the single conceptual shift on which the entire thesis rests: moving the estimation problem from the space of the uncertainty to the space of the constraint residual. The chapter closes by fixing the scope, the anchor literature, and the research questions of the thesis.

1.1 Where Deterministic Optimization Falls Short

This section establishes why the deterministic template struggles under uncertainty. It first identifies the mechanism behind the failure, then illustrates it across four application domains, works through a numerical example in which the failure can be quantified exactly, and closes by showing that the observed failure rate is largely independent of the application.

1.1.1 The mean-value approach and the flaw of averages

The standard nonlinear program

$$\underset{x \in X}{\text{minimize}}\ f(x) \quad \text{subject to} \quad g(x, \xi) \le 0, \tag{1}$$

treats the parameter vector ξ as a known constant. Decades of solver development have made problems of the form Eq. (1) routinely solvable at scale, and this success has had a quiet side effect: the deterministic template is applied by default, even when ξ is plainly a random quantity. The usual workaround is to replace ξ with a point estimate, most often its mean or a nominal forecast, and solve the resulting deterministic problem. This is the mean-value approach, and its shortcomings are systematic rather than occasional.

The failure mechanism is worth stating carefully, because it is the mechanism behind the examples in this section. An optimizer pushes its solution to the boundary of the feasible region; that is what optimality means whenever a constraint is binding. If the constraint $g(x, \bar{\xi}) \leq 0$ is active at the mean-value optimum $x^\star_{\mathrm{mv}}$, then the residual $g(x^\star_{\mathrm{mv}}, \xi)$ sits exactly at zero when ξ equals its mean. The moment ξ is allowed to fluctuate around that mean, roughly half of its realizations push the residual positive and the constraint is violated. The optimizer does not merely tolerate risk; it tends to select the configuration in which the risk of violation is largest, because that configuration is where the objective is best. In this sense optimization and uncertainty often work against each other, a phenomenon known in the planning literature as the flaw of averages (Savage, 2009): a plan built on average inputs need not behave like an average plan, and can be considerably more fragile.

1.1.2 Four domains where uncertainty bites

Four application domains make this concrete. They are chosen deliberately, because each reappears later in the thesis as a benchmark or a case study, and because together they span the distributional pathologies that motivate the nonparametric approach of Section 1.4.

Autonomous lunar landing. A powered-descent guidance problem asks for a thrust profile that brings a lander from orbit to a designated site while respecting bounds on velocity, glide slope, and fuel. The dynamics are integrated forward under thrust perturbations and imperfectly known gravitational and terrain parameters. Keil u. a. (2021) show that disturbance distributions in this setting are naturally non-Gaussian; a mixture arises, for example, when a thruster operates in one of two regimes. A trajectory optimized for nominal thrust grazes the velocity constraint at touchdown, so under perturbation the lander exceeds its safe impact speed on a large fraction of descents. A lunar landing offers no opportunity to repeat a failed attempt. Chance-constrained formulations of exactly this problem are the application anchor of the thesis (Keil u. a., 2020, 2021; Blackmore u. a., 2010).

Energy dispatch under renewable generation. A system operator commits generation a day ahead against uncertain demand and uncertain wind and solar output. Forecast errors for renewable infeed are well documented to be skewed and heavy-tailed rather than Gaussian (Hodge u. Milligan, 2011; Bienstock u. a., 2014). A dispatch that balances the forecast exactly is short of energy whenever net demand lands above its forecast, which by construction happens with probability near one half, and reserve activations, load shedding, or price spikes follow. This domain has produced some of the strongest data-driven chance-constraint literature (Ciftci u. a., 2019; Wu u. a., 2021; Wu u. Kargarian, 2023).

Chemical process control. A reactor recipe is optimized for yield subject to temperature and purity limits, but the composition of the feed material varies from batch to batch. Running the process at the deterministic optimum places the temperature constraint on the boundary, and ordinary feedstock variation then produces off-specification batches at a steady rate. Process systems engineering was, in fact, the field where data-driven kernel-smoothed chance constraints were first formulated (Calfa u. a., 2015).

Portfolio allocation. An investor maximizes expected return subject to a cap on the probability of a large loss. Asset returns exhibit heavy tails and volatility clustering; these are among the most robust stylized facts in empirical finance (Cont, 2001). A portfolio optimized as if returns were fixed at their historical means concentrates in the highest-mean assets and breaches its loss cap far more often than any Gaussian calculation predicts (Rockafellar u. Uryasev, 2000; Liu u. a., 2022).

1.1.3 A worked example: the cost of trusting the mean

The mechanism is easiest to see in a problem small enough to solve by inspection. Consider a single-bus dispatch model. A controllable resource of capacity x megawatts must cover an uncertain net demand ξ, and capacity is costly, so the planner solves

$$\underset{x\geq 0}{\text{minimize}}\; c\,x \quad \text{subject to} \quad \xi - x \leq 0. \tag{2}$$

Let net demand follow a moderately skewed lognormal law with mean $\bar{\xi} \approx 102\,\text{MW}$, a realistic shape for net load once renewable infeed is subtracted (Hodge u. Milligan, 2011). The mean-value method replaces ξ by $\bar{\xi}$ and returns $x^\star_{\text{mv}} = \bar{\xi}$: build exactly

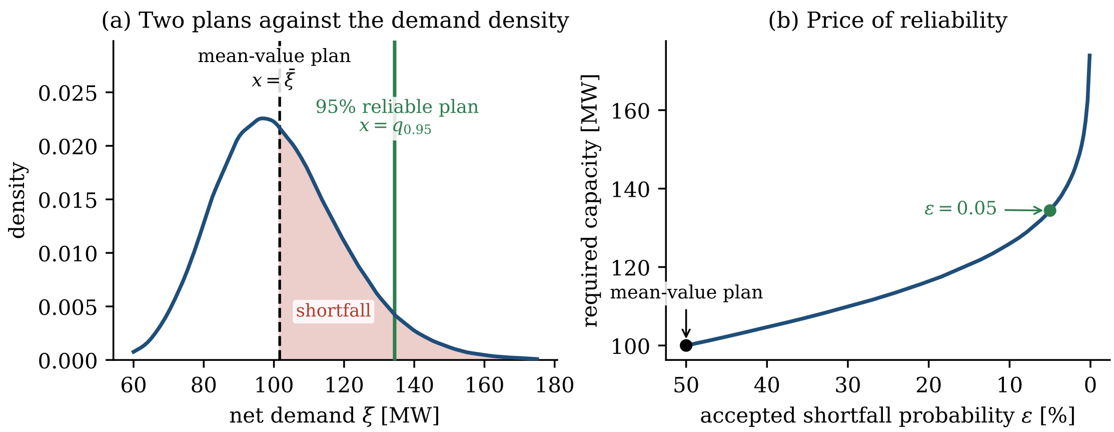


Figure 1: The mean-value fallacy in a one-dimensional dispatch problem with lognormal net demand (Monte Carlo with 2×10^5 samples). (a) The mean-value plan $x = \bar{\xi} \approx 102\,\text{MW}$ leaves the shaded shortfall region uncovered, so the constraint $\xi \leq x$ fails on 46% of days; covering demand with 95% reliability requires $x = q_{0.95} \approx 134\,\text{MW}$. (b) Required capacity as a function of the accepted shortfall probability ε. Reliability is bought at an accelerating price as $\varepsilon \to 0$; the chance constraint makes this price explicit and lets the planner choose a point on the curve deliberately.

average capacity. Figure 1(a) shows what this decision looks like against the actual demand density. The constraint is violated on every day that demand exceeds its mean, and for this distribution that is 46% of all days. A plan that appears exact on paper fails almost every other day in operation.

The repair is equally easy to read off the figure. If the planner is willing to accept a shortfall on at most 5% of days, the appropriate decision is the 95% quantile of demand, $x = q_{0.95} \approx 134\,\text{MW}$, about 32% more capacity than the mean-value plan. Figure 1(b) generalizes this observation into the central trade-off of the thesis: required capacity, and hence cost, rises steeply as the accepted violation probability ε shrinks, and rises without bound as $\varepsilon \to 0$ because the lognormal has unbounded support. Reliability is a commodity with a price, and the role of a properly formulated stochastic optimization problem is to make that price visible and chooseable rather than implicit and accidental.

1.1.4 A failure rate that recurs across problems

One might hope that the 46% figure is an artifact of the simple model. It is not. Figure 2 repeats the experiment for stylized versions of three of the domains above: dispatch under skewed demand, a landing-corridor constraint under bimodal thrust

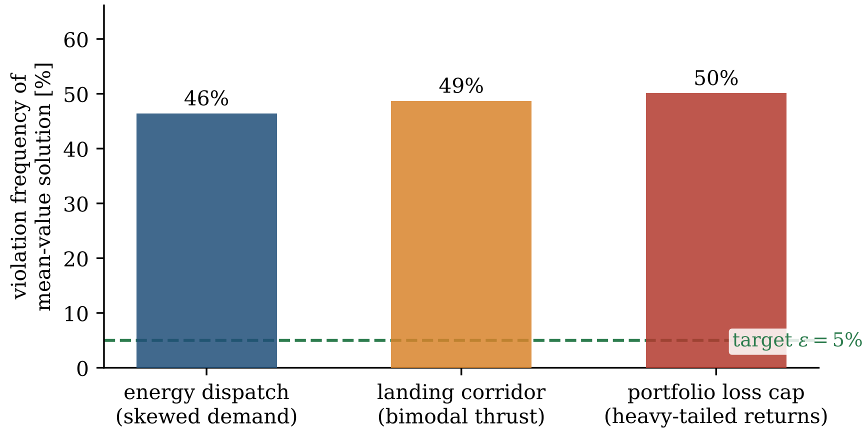


Figure 2: Empirical violation frequency of the mean-value solution in three stylized problems drawn from the application domains of this thesis (Monte Carlo estimates). A typical engineering target of $\varepsilon = 5\%$ is shown for scale. The recurring value close to 50% reflects the boundary mechanism: optimizers drive binding constraints to the edge of feasibility, where roughly half of all perturbations cause failure.

disturbance, and a portfolio loss cap under heavy-tailed (Student-t) returns. In every case the mean-value solution violates its constraint between 46% and 50% of the time, well beyond a typical design target. The violation frequency stays near one half largely because of the boundary mechanism described above: where the constraint is binding at the deterministic optimum, the median of the residual sits at or near zero, and this holds across the applications considered. The value therefore reflects a property of the optimization itself rather than of any particular application domain.

1.2 Two Extremes and the Gap Between Them

This section examines the two classical responses to the failure just described, the deterministic status quo and worst-case robustness, identifies the settings in which each is appropriate, and introduces the chance constraint as a formulation that connects the two through a tunable reliability level.

1.2.1 Ignoring uncertainty and worst-case robustness

Once the limitations of the mean-value approach are recognized, the question becomes what should replace it. Two answers dominate, and they sit at opposite ends of a spectrum.

The first answer is to ignore uncertainty entirely, which is the mean-value approach already examined in Section 1.1. Its defining property is worth restating in the language used for the rest of the thesis: the deterministic solution carries essentially no reliability statement. The formulation Eq. (1) does not bound the probability of failure, and Fig. 2 shows that this probability is typically near one half. Whatever reliability a deployed mean-value solution exhibits usually comes from safety margins added by hand, outside the optimization. Such hand-tuned margins can be read as an indication that the optimization, as posed, did not capture the intended requirement.

The second answer is worst-case robust optimization (Soyster, 1973; Ben-Tal u. a., 2009). The uncertain parameter is confined to an uncertainty set Ξ, and the constraint is enforced for every element of that set:

$$g(x, \xi) \leq 0 \quad \text{for all } \xi \in \Xi. \tag{3}$$

Robust optimization is a mature and elegant theory; for many set geometries, Eq. (3) admits exact tractable reformulations, and when a failure is not acceptable under any circumstance it is arguably the model of choice. Its weakness is the obverse of its strength: protection is purchased against every point of Ξ, including points that are vanishingly unlikely or physically implausible, and the objective pays for all of them. In the dispatch example, the demand distribution has unbounded support, so a literal worst-case formulation has no feasible solution at all; the set Ξ must be truncated by fiat, and the answer then depends sensitively on a truncation the data does not specify. Even with a defensible truncation, the protection level is largely a fixed property of the set rather than a dial the decision maker controls. Empirically, robust solutions in power systems and aerospace routinely sacrifice a large fraction of attainable objective value to guard against scenarios whose combined probability is far below any operational concern (Ben-Tal u. a., 2009; Keil u. a., 2021).

1.2.2 When each extreme is the right tool

Fairness demands an acknowledgment before the middle ground is claimed: each extreme has a legitimate home. The mean-value approach is correct when uncertainty is genuinely negligible relative to the constraint margins, or when violations carry only a linear cost that the objective already prices in, as in inventory models where a shortfall is simply an expense. Worst-case robustness is correct when the uncertainty set is small, well characterized, and physically bounded, and when a single failure would be severe and difficult to reverse: structural safety factors are a familiar example. The

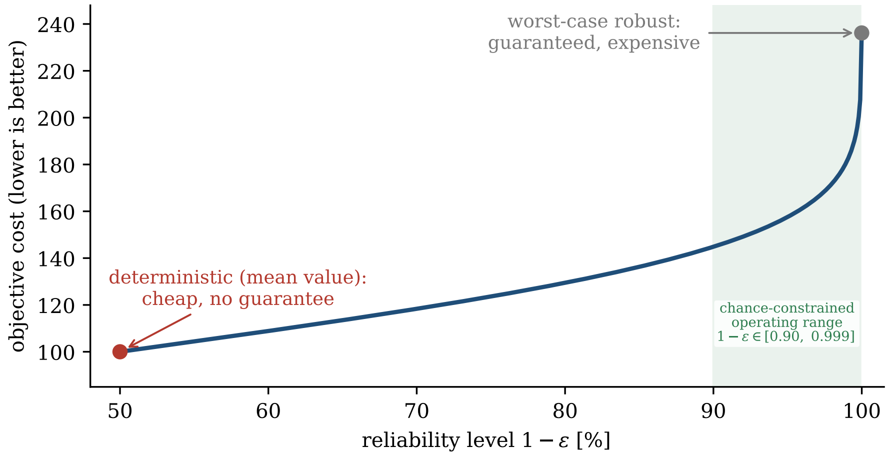


Figure 3: The reliability spectrum for the dispatch example of Fig. 1. The deterministic mean-value plan is cheap and offers no guarantee; the worst-case robust plan is guaranteed and expensive, and is not even well defined under unbounded support. Chance constraints parameterize the entire curve between the extremes: the decision maker selects a reliability level $1-\varepsilon$, and the optimization returns the cheapest decision that honors it.

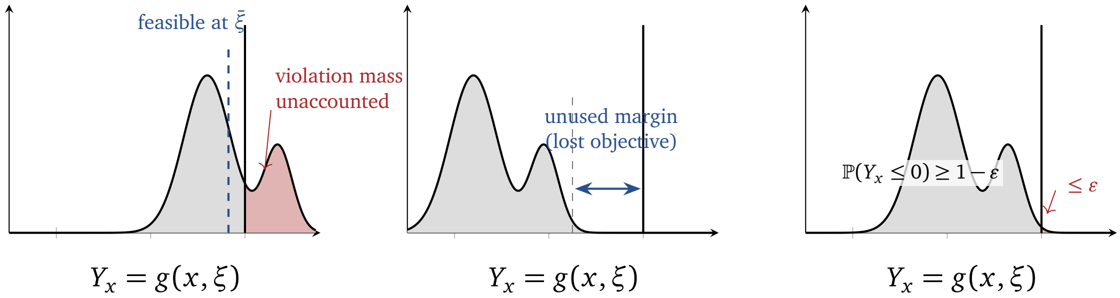
problems this thesis targets sit in neither home. Their uncertainties are large, data-described, and unbounded or effectively so, and their failures are costly but tolerable at a small, explicit rate. A lander program accepts a quantified risk per descent; a grid code tolerates a bounded loss-of-load expectation per year. For this regime, neither extreme expresses what the decision maker actually wants.

1.2.3 Chance constraints as the unifying middle ground

Between a method with no guarantee and a method that guarantees too much lies the formulation this thesis is about. A chance constraint, introduced by Charnes u. Cooper (1959) and developed into a full branch of stochastic programming by Prékopa (1995), requires the constraint to hold with a prescribed probability:

$$\mathbb{P}\big(g(x,\xi) \le 0\big) \ge 1-\varepsilon. \tag{4}$$

The risk tolerance $\varepsilon \in (0,1)$ is small, typically between 0.001 and 0.1 depending on the stakes, and $1-\varepsilon$ is the reliability level. Parts of the literature, including the convergence theory of Schuster (2025), write the required satisfaction probability directly as $\alpha = 1-\varepsilon$; the two conventions are interchangeable, and this thesis works with ε throughout. Figure 3 places the three formulations on a single axis for the dispatch example. Rather than a compromise between the extremes, the chance constraint can



(a) Mean-value
optimistic, risk unquantified

(b) Worst-case robust
safe, conservative

(c) Chance-constrained
risk explicit: ε chosen

Figure 4: Three responses to constraint uncertainty, shown on the density of the residual $Y_x = g(x, \xi)$; the constraint is satisfied for $Y_x \leq 0$. **(a)** The mean-value solution satisfies the constraint at $\bar{\xi}$ but leaves an unquantified violation mass (red). **(b)** The worst-case robust solution eliminates all violations by pushing the distribution well clear of zero, at the cost of large unused margin. **(c)** The chance-constrained solution places exactly the permitted fraction ε of probability mass beyond the boundary, making the accepted risk explicit and controllable.

be viewed as generalizing both of them. As $\varepsilon \to 0$ it approaches worst-case protection over the support of ξ, and at $\varepsilon = 0.5$ it roughly reproduces the mean-value behavior for symmetric residuals. Every intermediate point is available, and each carries an explicit, auditable meaning: the system fails at most a fraction ε of the time. This closely mirrors the language in which engineering requirements are commonly written. Grid codes prescribe loss-of-load expectations, flight rules prescribe acceptable failure probabilities per descent, and risk mandates prescribe loss probabilities per quarter. The chance constraint is the optimization-native translation of a reliability requirement, which is why it is a natural framing for many engineering and operational problems, and why the difficulty of solving it, the subject of Section 1.3, has remained an active research frontier for over six decades. Figure 4 makes the three positions concrete. Each panel shows the same bimodal residual density $Y_x = g(x, \xi)$, with the constraint boundary at zero; the two modes reflect the non-Gaussian disturbance shapes that motivate the nonparametric approach developed from subsection 1.4 onward

1.3 What Chance Constraints Are and Why They Are Hard

This section fixes the notation used throughout the thesis, formalizes individual and joint chance constraints, and isolates the three structural properties that make probabilistic constraints computationally demanding. These three difficulties also organize the survey of solution methods in Chapter 2.

1.3.1 Individual and joint chance constraints

This section fixes notation used throughout the thesis and isolates the three structural sources of computational difficulty. Let $x \in X \subseteq \mathbb{R}^n$ be the decision vector, let ξ be a random vector on a probability space $(\Omega, \mathscr{F}, \mathbb{P})$ taking values in $\mathbb{R}^d$, and let $g : \mathbb{R}^n \times \mathbb{R}^d \to \mathbb{R}$ be a constraint function. The *individual chance constraint* is

$$\mathbb{P}\big(g(x,\xi) \le 0\big) \ge 1 - \varepsilon. \tag{5}$$

When the system carries several uncertain requirements $g_1, \ldots, g_R$ that must hold simultaneously, the appropriate model is the *joint chance constraint*

$$\mathbb{P}\big(g_r(x,\xi) \le 0,\ r = 1,\ldots,R\big) \ge 1 - \varepsilon, \tag{6}$$

which bounds the probability that *all* requirements hold at once. The distinction matters: enforcing each g_r individually at level $1 - \varepsilon$ does not bound the joint failure probability, while the classical repair via Boole's inequality, splitting the budget as $\sum_r \varepsilon_r \le \varepsilon$, is sound but conservative (Prékopa, 1995; Keil u. a., 2021). The full chance-constrained program studied in this thesis is

$$\underset{x \in X}{\text{minimize}}\ f(x) \quad \text{subject to} \quad \mathbb{P}\big(g(x,\xi) \le 0\big) \ge 1 - \varepsilon. \tag{7}$$

A word on the role of ε is warranted, because it is the one quantity in Eq. (7) that mathematics cannot supply. The risk tolerance is a design parameter set by the consequences of failure and by external requirements: a 5% daily shortfall tolerance may be acceptable for a microgrid with backup supply (Ciftci u. a., 2019), while a descent constraint may demand $\varepsilon \le 10^{-2}$ or far less (Blackmore u. a., 2010). Two facts about this parameter shape everything downstream. First, the difficulty of the problem grows as ε shrinks, because the constraint then probes ever deeper into the tail of a distribution that must be known or estimated ever more accurately there. Second, ε is a budget, and for joint constraints Eq. (6) the budget must cover all R requirements simultaneously, which is what makes the joint form strictly harder than the individual one. Throughout the thesis, ε is treated as given and the question is how to honor it efficiently and honestly.

1.3.2 Three structural sources of difficulty

Writing Eq. (7) is easy; solving it is not, and the obstruction is threefold. Each difficulty below is independent of the others, and any viable solution method must answer all three. The pattern repeats through the entire methodological literature, so these three points are the lens through which every method in this thesis, including the KDE reformulation itself, should be judged.

Difficulty 1: the feasible set is generally non-convex. Convexity of g in x does not transfer to the probabilistic feasible set $\{x : \mathbb{P}(g(x,\xi) \le 0) \ge 1-\varepsilon\}$. A minimal example makes the geometry vivid. Take a linear constraint $\xi^\top x \le 1$ where ξ takes one of two values $\xi^{(1)}, \xi^{(2)}$ with probability one half each, and set $\varepsilon = 0.5$. The chance constraint then requires $\xi^\top x \le 1$ for *at least one* of the two scenarios, so the feasible set is the union of two half-spaces. Figure 5(a) shows the result: two points, each satisfying a different scenario, are individually feasible, while their midpoint satisfies neither and is infeasible. A union of convex sets is not convex, and this toy construction scales up: with general discrete or continuous distributions the feasible set can shatter into combinatorially many pieces. Convexity survives only under special structure, most famously when the distribution of ξ is log-concave and g has suitable form, the well-known theory of Prékopa (1995), or when ξ is Gaussian and the constraint is linear in ξ, in which case Eq. (5) collapses to a second-order cone constraint (Charnes u. Cooper, 1963; Shapiro u. a., 2014). Outside these islands, non-convexity is the rule, and global optimality claims must be abandoned in favor of high-quality local solutions, which is precisely the regime in which nonlinear programming solvers operate.

Difficulty 2: the probability has no closed form. Define the parameterized failure region $M(x) = \{\xi : g(x,\xi) > 0\}$. Evaluating the constraint in Eq. (7) means computing

$$\mathbb{P}\big(g(x,\xi) \le 0\big) = 1 - \int_{M(x)} \rho_\xi(\xi)\, d\xi, \tag{8}$$

an integral of the density ρ_ξ over a region whose shape changes with every candidate decision x. For nonlinear g the region $M(x)$ is curved and possibly disconnected, the integral is multidimensional, and a closed form exists only in textbook cases. Worse, in the data-driven settings that motivate this thesis the density ρ_ξ itself is unknown; all that exists is a finite collection of samples from simulation, historical records, or exper-

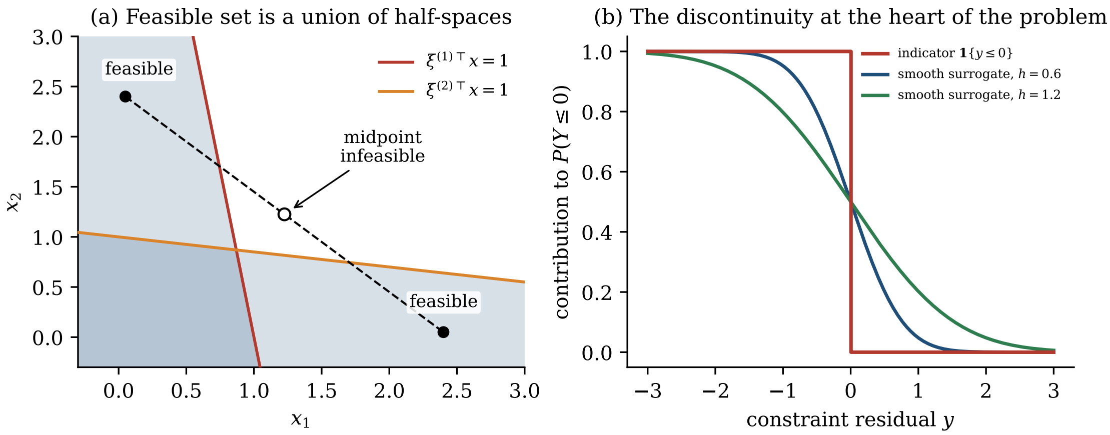


Figure 5: The two geometric faces of chance-constraint hardness. (a) For a two-point distribution of ξ and $\varepsilon = 0.5$, the feasible set of $\mathbb{P}(\xi^\top x \le 1) \ge 0.5$ is the union of two half-spaces (shaded). The marked endpoints are feasible while their midpoint is not, certifying nonconvexity. (b) The contribution of a single sample to the probability is the discontinuous indicator $\mathbf{1}\{y \le 0\}$ of the residual; its derivative is zero almost everywhere and undefined at the jump. Smooth surrogates obtained by integrating a kernel, shown for two bandwidths h, restore the differentiability that gradient-based solvers require, at the price of a bias controlled by h. This single picture contains the central tension of the thesis.

iments. The constraint function that the optimizer must evaluate, thousands of times, at thousands of points x, is therefore an unknown integral of an unknown density over an intractable region. Any practical method replaces Eq. (8) with an approximation, and the quality of that approximation governs everything downstream.

Difficulty 3: the probability is built from a discontinuous function. Rewriting the probability as an expectation,

$$\mathbb{P}\big(g(x,\xi) \le 0\big) = \mathbb{E}\big[\, \mathbf{1}\{g(x,\xi) \le 0\}\, \big], \tag{9}$$

exposes the third obstruction: the indicator $\mathbf{1}\{\cdot\}$ jumps from one to zero at the constraint boundary, as plotted in Fig. 5(b). The natural sample-based estimate of Eq. (9), count the fraction of samples satisfying the constraint, inherits this discontinuity. As a function of x it is piecewise constant: its gradient is zero almost everywhere and undefined on a measure-zero set, which is poorly suited to the sequential quadratic programming and interior-point methods that are widely used in nonlinear optimization (Shapiro u. a., 2014). The established escape routes each pay a recognizable price. Mixed-integer reformulations of the counting constraint are exact but combinatorial and scale poorly (Luedtke u. Ahmed, 2008). The scenario approach enforces the constraint on all N sampled scenarios, regaining convexity and even finite-sample feasibility certificates for convex problems (Calafiore u. Campi, 2006; Campi u. Garatti, 2008), but it cannot exploit the slack that ε permits and grows conservative as N increases. Conservative convex surrogates such as conditional value-at-risk restrict the feasible set from inside (Rockafellar u. Uryasev, 2000; Nemirovski u. Shapiro, 2006), guaranteeing feasibility at a cost in objective value. Smoothing methods, the family to which this thesis belongs, replace the indicator with a differentiable approximation (Peña-Ordieres u. a., 2020), restoring the gradients that nonlinear solvers need while introducing a controlled bias, visible in Fig. 5(b) as the gap between the step and the smooth curves.

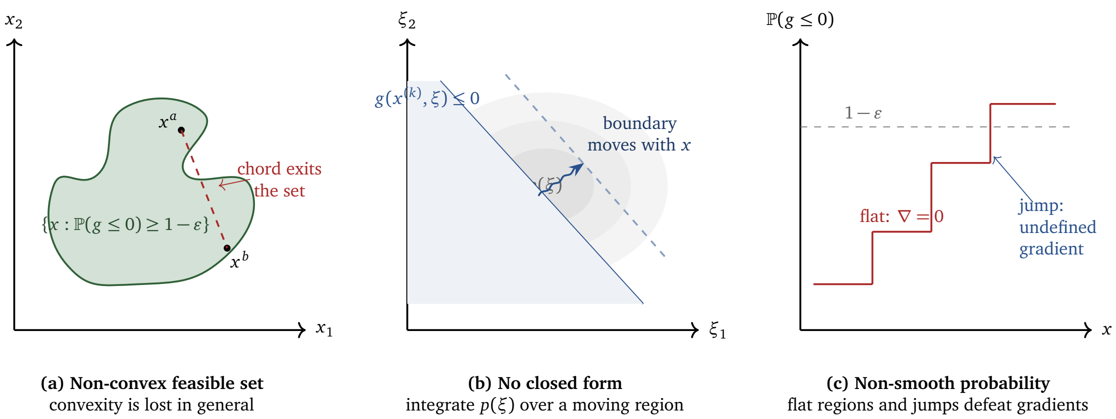


Figure 6: The three structural sources of difficulty in chance-constrained programming: **(a)** the feasible set is non-convex, **(b)** the constraint probability has no closed form because the safe region moves with x, and **(c)** the probability function is flat almost everywhere and non-differentiable at the boundary.

The combination of the three difficulties, and not any one of them alone, is what makes Eq. (7) substantially harder than a deterministic NLP. Non-convexity generally precludes global guarantees, the missing closed form precludes exact evaluation, and non-smoothness makes naive sample averaging unsuitable inside derivative-based solvers. Every approximation method discussed in this thesis is an attempt to buy back tractability by conceding something: exactness, generality, or a margin of conservatism. The contribution of the KDE-based reformulation is a specific and unusually favorable bargain on all three fronts, and the remainder of this chapter explains why.

1.4 The Case for Nonparametric Estimation

This section develops the argument for estimating distributions from data. It reviews the parametric alternative and the evidence against its default assumptions, quantifies the consequences of a misspecified distribution, introduces kernel density estimation, and closes with an account of the costs that the nonparametric route incurs.

1.4.1 The parametric assumption and its routine failure

Among the classical escape routes from Section 1.3, the oldest and still most widely used is parametric: assume the law of ξ belongs to a named family, almost always the Gaussian, and exploit the structure that follows. Under Gaussian ξ and a constraint linear in ξ, the chance constraint Eq. (5) reduces exactly to a deterministic second-order cone constraint involving the mean, the covariance, and a quantile of the standard normal (Charnes u. Cooper, 1963; Shapiro u. a., 2014). The reformulation is convex, smooth, and supported by every modern conic solver. When the Gaussian assumption holds, this reformulation is difficult to improve upon, and the experimental chapters use it as the reference baseline for that reason.

The assumption, however, is routinely false in the very domains where chance constraints matter most, and the evidence is not anecdotal. Wind-power forecast errors are skewed and heavy-tailed, with documented deviations from normality that grow in the tails, which is precisely where a chance constraint lives (Hodge u. Milligan, 2011; Wan u. a., 2020). Equity and commodity returns exhibit heavy tails and aggregational Gaussianity only at long horizons; at decision-relevant horizons their kurtosis is far above three (Cont, 2001). Thrust disturbances and navigation errors in powered descent arise from mixtures of operating regimes and are naturally multimodal (Keil u. a., 2021). Process feedstocks vary by supplier and batch in ways no single unimodal family captures (Calfa u. a., 2015). The Gaussian is often chosen more for analytical convenience than for its fit to the observed behavior of the system.

1.4.2 What a wrong distribution does to the guarantee

The consequences of the mismatch are asymmetric and quantifiable. Figure 7 dissects both failure modes. Panel (a) shows a bimodal disturbance law next to its moment-matched Gaussian fit: the fit invents probability where there is none and erases the two modes that actually govern system behavior. Panel (b) shows the more dangerous

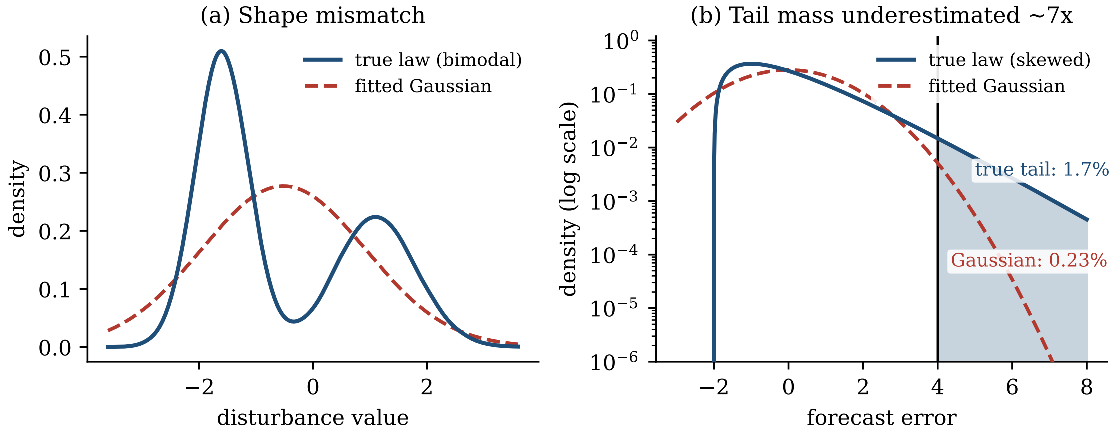


Figure 7: What a wrong distributional assumption does to a chance constraint. (a) A bimodal disturbance law, of the kind produced by two-regime actuator behavior, against its best Gaussian fit: the fit places probability mass between the modes where the true law has almost none, and misplaces both shoulders. (b) A skewed forecast-error law (centered Gamma) on a logarithmic density scale. Beyond the marked threshold the Gaussian fit reports 0.23% tail mass where the true law carries 1.7%, an underestimate by a factor of about seven. A planner who certifies a 0.5% risk level with the Gaussian model deploys a system whose true failure rate is more than three times the certificate.

case on a logarithmic scale. For a skewed forecast-error law, the Gaussian fit underestimates the mass beyond a critical threshold by a factor of seven: it reports 0.23% where the truth is 1.7%. A decision certified to a 0.5% failure probability under the Gaussian model in fact fails more than three times as often as certified. The error can also point the other way: when the true law is lighter-tailed than the fit, the parametric constraint is needlessly conservative and pays an invisible objective penalty. Whether the result is optimistic or conservative, the direction is governed by distributional geometry that the parametric fit does not capture, and the certificate that the chance constraint was intended to provide, $\mathbb{P}(\text{failure}) \leq \varepsilon$, may no longer hold. A reliability guarantee that rests on an unverified distributional assumption therefore provides considerably weaker assurance than its form suggests.

1.4.3 Kernel density estimation and its fit to optimization

The alternative pursued in this thesis is to estimate the distribution from the data instead of assuming it. Kernel density estimation, the canonical nonparametric estimator introduced by Rosenblatt (1956) and Parzen (1962), reconstructs a density from samples $y_1, \dots, y_N$ as

$$\hat{\rho}_h(y) = \frac{1}{Nh} \sum_{j=1}^{N} K\left(\frac{y - y_j}{h}\right), \tag{10}$$

where the kernel K is a smooth density and the bandwidth $h > 0$ sets the degree of smoothing. The estimator makes no shape commitment: given enough data it recovers skewness, multimodality, and heavy tails alike, with well-understood consistency theory and practical bandwidth rules (Silverman, 1986; Devroye u. Györfi, 1985). Three properties make KDE specifically suited to the optimization context, rather than merely to statistics. First, $\hat{\rho}_h$ is as smooth as its kernel, so quantities built from it are differentiable, which addresses Difficulty 3 head-on. Second, its integral is available in closed form whenever the kernel's cumulative distribution function is, which addresses the evaluation half of Difficulty 2. Third, it is distribution-free, which removes the implicit distributional assumption discussed above.

This is not a speculative combination. A coherent body of work already demonstrates KDE-based chance constraints across the application domains of Section 1.1: kernel-smoothed individual and joint chance constraints in process planning (Calfa u. a., 2015), KDE-reformulated optimal control for launcher ascent and the Goddard problem (Caillau u. a., 2018), biased-KDE optimal control for soft lunar landing (Keil u. a., 2020, 2021; Keil u. Rao, 2021), adaptive-KDE microgrid energy management (Ciftci, 2017; Ciftci u. a., 2019; Babić u. a., 2023), multivariate-KDE joint chance constraints for economic dispatch and power-system scheduling (Wu u. a., 2021; Wu u. Kargarian, 2023), KDE-driven distributionally robust load shedding (Qi u. a., 2026), and KDE-based tail-risk portfolio optimization (Liu u. a., 2022). Adjacent nonparametric formulations reinforce the same direction in robotics and forecasting (Gopalakrishnan u. a., 2018; Wan u. a., 2020; Wang u. a., 2021). What this literature has lacked, until recently, is a unifying account connecting the engineering constructions to convergence theory; supplying and stress-testing that connection is the work of this thesis.

1.4.4 The price of going nonparametric

A nonparametric estimator does not come free, and the costs are named now because they structure the experimental program of the application chapter. KDE suffers the curse of dimensionality: its convergence rate degrades rapidly as the dimension of the estimated variable grows (Silverman, 1986). The bandwidth h is a genuine hyperparameter whose role inside an optimization problem, where it shapes the feasible region and the gradients the solver sees, is more delicate than its role in density estimation. A plain KDE smooths in both directions, so it can overestimate the safe probability and thereby weaken the very bound it is meant to enforce. A fourth cost concerns the tails: a chance constraint with a small violation budget asks about a rare event, and a kernel estimate is least accurate in sparsely sampled tail regions. The severity of this effect is problem dependent rather than universal, since at moderate risk levels the binding region lies in the bulk of the distribution, where the estimator behaves well, while stringent levels push it into the tails. Each of these costs has a specific answer in this thesis: the first is mitigated by the residual-space construction of Section 1.5, the second is treated as a first-class experimental question, the third is addressed by the biased-KDE construction of Keil u. a. (2021), and the fourth is handled by reporting results across a range of risk levels.

1.5 The Residual-Space Insight

This section introduces the change of variables on which the methodological part of the thesis is built. It defines the constraint residual, explains why estimating its distribution is preferable to estimating the distribution of the uncertainty itself, and previews the treatment of joint constraints together with the role of the bandwidth.

1.5.1 From uncertainty space to residual space

Everything assembled so far points toward a single conceptual move, simple to state and central to the approach developed in this thesis. The naive way to use KDE in Eq. (7) is to estimate the density of the uncertainty ξ itself and then integrate the estimate over the safe region $\mathbb{R}^d \setminus M(x)$. This inherits both halves of Difficulty 2: the estimation runs in d dimensions, exactly where the curse of dimensionality bites, and the integration region still deforms with every candidate x. A more effective object to estimate is not the uncertainty itself but the *constraint residual*

$$Y(x) = g(x, \xi), \tag{11}$$

the scalar random variable obtained by pushing the uncertainty through the constraint function at the current decision. The residual is the quantity the constraint actually speaks about: $Y(x) \leq 0$ *is* constraint satisfaction, by definition, and the chance con-

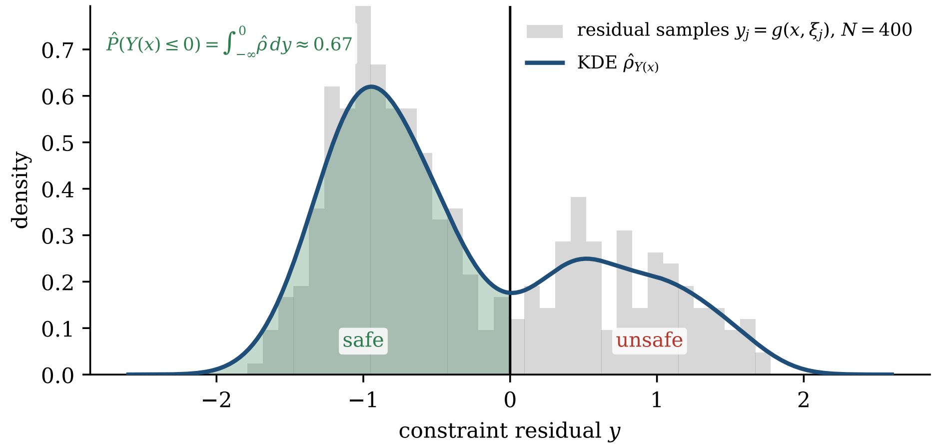


Figure 8: The residual-space construction at a fixed candidate decision x. Samples ξ_j of the uncertainty are pushed through the constraint to give residual samples $y_j = g(x, \xi_j)$ (histogram, $N = 400$, bimodal by construction); a KDE $\hat{\rho}_{Y(x)}$ smooths them, and its integral over the fixed safe region $(-\infty, 0]$ (shaded) estimates the safe probability, here $\hat{P} \approx 0.67$. The estimation problem is one-dimensional regardless of the dimension of ξ, and the resulting constraint $\hat{F}_{Y(x)}(0) \geq 1 - \varepsilon$ is a smooth, differentiable function of x.

straint Eq. (5) is nothing but a statement about the cumulative distribution function of $Y(x)$ at zero,

$$\mathbb{P}\big(Y(x) \leq 0\big) = F_{Y(x)}(0) \geq 1 - \varepsilon. \tag{12}$$

Schuster (2025) makes this change of variables the foundation of the convergence theory: the probability is rewritten as an integral of the density of the *transformed* variable $g(x, \xi)$ over the fixed set $(-\infty, 0]$, rather than an integral of the density of ξ over the moving set $\mathbb{R}^d \setminus M(x)$.

1.5.2 Three dividends of the change of variables

The benefits of this shift are worth enumerating, because each one addresses a specific difficulty from Section 1.3.

First, *dimensionality collapses*. The uncertainty ξ may live in ten or a hundred dimensions; the residual of an individual constraint is a scalar, always, and the residuals of a joint constraint form a vector of dimension R, the number of constraint components, which in the problems of this thesis is small. This is where the *curse of dimensionality*, named as the first cost of nonparametric estimation in Section 1.4.4, is addressed. The convergence rate of a kernel estimator deteriorates rapidly with the dimension of the estimated variable, so that holding the accuracy fixed requires a sample size that grows roughly exponentially in that dimension (Silverman, 1986). The decisive observation is that the relevant dimension for the residual-space construction is the dimension of the estimated variable, not of the uncertainty: KDE is asked to perform one-dimensional density estimation from a few hundred samples, the task it is best suited to, and the curse in d is largely avoided by construction. The constraint function g itself performs the dimension reduction, and it does so without information loss for the question being asked, since Eq. (12) is an identity, not an approximation. The mitigation is not unconditional, however; it weakens when many constraints must hold jointly, since the residual vector then carries the dimension R, a point taken up in Section 1.5.3.

Second, *the integration region freezes*. The safe region in residual space is $(-\infty, 0]$ for every decision x; all dependence on x has migrated into the samples $y_j = g(x, \xi_j)$ themselves. Integrating the kernel estimate over a fixed half-line is elementary: substituting Eq. (10) gives the estimated CDF

$$\hat{F}_{Y(x)}(0) = \frac{1}{N}\sum_{j=1}^{N} \mathscr{K}\left(\frac{-g(x,\xi_j)}{h}\right), \qquad \mathscr{K}(t) = \int_{-\infty}^{t} K(s)\,ds, \tag{13}$$

a finite sum of evaluations of the kernel's CDF $\mathscr{K}$, available in closed form for every kernel used in this thesis.

Third, *smoothness is restored*. Comparing Eq. (13) with the sample-average form Eq. (9) reveals the substitution at the heart of the method: the discontinuous indicator $\mathbf{1}\{g(x,\xi_j) \le 0\}$ has been replaced by the smooth sigmoid $\mathscr{K}(-g(x,\xi_j)/h)$, exactly the curves drawn in Fig. 5(b). If g is differentiable in x, then so is $\hat{F}_{Y(x)}(0)$, with gradients obtained by the chain rule through $\mathscr{K}$. The chance constraint Eq. (5) thereby becomes the deterministic, smooth, nonlinear constraint

$$\hat{F}_{Y(x)}(0) \ge 1 - \varepsilon, \tag{14}$$

which any sequential quadratic programming, interior-point, or direct-collocation solver consumes natively. Figure 8 shows the whole construction in one frame: residual samples, the kernel estimate, the frozen safe region, and the resulting probability estimate.

1.5.3 Joint constraints, bandwidth, and the safety guarantee

The construction extends to joint chance constraints with one further idea that previews the extension studied in this thesis. For Eq. (6), the natural scalar summary of the constraint system is the worst violation, $Y(x) = \max_r g_r(x,\xi)$, since all requirements hold exactly when the maximum residual is nonpositive. The maximum preserves the one-dimensionality of the estimation problem but reintroduces a kink; smoothing it with a log-sum-exp surrogate restores differentiability at the cost of a second, controllable approximation layer. This scalar violation statistic, its interaction with the KDE layer, and the alternative of low-dimensional multivariate KDE over the residual vector form the substance of research question RQ4 and are developed in Part 3.

The reformulation Eq. (14) is where the thesis begins rather than ends, because the two costs flagged in Section 1.4 are still standing and they define the methodological arc of Part 3. The bandwidth h now controls the geometry of an optimization problem, not just the fidelity of a density plot: $h \to 0$ recovers the non-smooth indicator and its numerical pathologies, while large h oversmooths and distorts the feasible region, and no classical bandwidth rule was derived with either effect in mind. And because an ordinary KDE can overestimate $F_{Y(x)}(0)$, the constraint Eq. (14) can certify a reliability the true system does not possess. The constructive answer is the biased KDE of Keil u. a. (2020, 2021): kernels engineered so that their integral function bounds the indicator from the safe side, making Eq. (14) a guaranteed conservative approximation of the true chance constraint while remaining smooth and bounded. The theoretical answer is the convergence framework of Schuster (2025), which establishes conditions under which solutions of the KDE-approximated problem converge to solutions of the true problem as $N \to \infty$. Engineering construction and convergence theory, developed independently, meet in residual space, and their meeting is the subject of this thesis.

1.6 Thesis Scope and Positioning

This closing section delimits the scope of the thesis. It positions the work between its two anchor papers, states which questions lie outside the scope, and formulates the four research questions together with a roadmap of the remaining chapters.

1.6.1 Disciplinary positioning and anchor literature

A thesis at the intersection of three fields must say plainly what it is and what it is not. This thesis is not a contribution to nonparametric statistics: it proves no new properties of kernel density estimators, and it takes the consistency theory of Parzen (1962); Silverman (1986); Devroye u. Györfi (1985) as given. It is not a contribution to control theory: it establishes no stability or recursive feasibility results for closed-loop predictive control. It is a data science and applied optimization contribution. Its object of study is the KDE-based reformulation pipeline itself, from samples through residuals through Eq. (14) into a numerical solver, and its method is to implement that pipeline rigorously, analyze its behavior, compare it against the established alternatives, and validate it across problem settings chosen to stress its assumptions. One further boundary deserves explicit mention. Every guarantee developed in this thesis is a statement about the distribution from which the samples were drawn. Whether such a guarantee transfers to the distribution encountered in deployment, when samples come from a simulator or from historical records that represent the target system imperfectly, is a separate question that this thesis does not address; distributionally robust formulations (**?**) are a natural direction for that question.

Two anchor papers bound the territory. Keil u. a. (2021), with the earlier preprint Keil u. a. (2020) and the computational follow-up Keil u. Rao (2021), supply the constructive, engineering side: biased kernels whose integral functions bound the indicator, a safety-preserving deterministic reformulation, and a demonstration on chance-constrained soft lunar landing solved by direct collocation. Schuster (2025) supplies the analytical side: convergence theorems guaranteeing that, under verifiable conditions on the kernel, the bandwidth, and the problem, solving the estimated problem is asymptotically equivalent to solving the true one. The first anchor gives the method its practical credibility, the second its mathematical license; the thesis builds the bridge between them and load-tests it experimentally. The surrounding evidence base spans aerospace (Caillau u. a., 2018), energy systems (Ciftci u. a., 2019; Wu u. a., 2021; Wu u. Kargarian, 2023; Qi u. a., 2026), process systems (Calfa u. a., 2015), and finance (Liu u. a., 2022), and the comparative baselines are the classical ones: scenario optimization (Calafiore u. Campi, 2006; Campi u. Garatti, 2008), sample average approximation (Luedtke u. Ahmed, 2008; Pagnoncelli u. a., 2009), parametric reformulation (Charnes u. Cooper, 1963), and convex risk surrogates (Rockafellar u. Uryasev, 2000; Nemirovski u. Shapiro, 2006). Further smooth alternatives exist for structured settings, notably polynomial chaos expansions (Xiu u. Karniadakis, 2002) and spheric-radial gradient formulae for Gaussian-like laws (**?**); a systematic benchmark against these lies outside the scope of this thesis and is noted as future work.

1.6.2 Research questions and roadmap

Four research questions organize everything that follows.

- **RQ1: formulation** how is a chance-constrained problem reformulated through KDE over constraint residuals, for individual and joint constraints, and under what conditions is the reformulation mathematically justified?
- **RQ2: numerical behavior** how does the KDE reformulation compare with deterministic, scenario-based, and parametric approaches in feasibility, objective value, conservatism, runtime, and sensitivity to sample size?
- **RQ3: bandwidth and smoothing** how do bandwidth and smoothing choices shape feasibility, conservatism, and solver performance, given that h acts on the optimization geometry and not only on estimation error?
- **RQ4: extension** can the framework be strengthened through one focused extension, specifically scalar violation statistics for joint chance constraints, optionally with log-sum-exp smoothing?

The argument now proceeds in three movements. Part 2 supplies the technical vocabulary: the taxonomy of optimization under uncertainty, the theory of kernel density estimation, and the elements of numerical optimal control. Part 3 develops the mathematics in the order a practitioner would discover it: the blackbox sample-counting approach and its failure inside gradient-based solvers, the smooth unbiased KDE reformulation, the biased construction that restores the safety guarantee, and the convergence theory that certifies the whole. Part 4 validates the framework experimentally, from controlled benchmarks through the lunar landing problem, and distills the results into practical guidance. The aim is a defensible answer to a question that each of the systems in Section 1.1 raises implicitly: how safe is safe enough, and what is the cheapest decision that reliably delivers it?

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Erlangen, den July 7, 2026 Hierher die Unterschrift

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